EMF & FARADAY'S LAW

Transformer EMF *stationary loop, B changes*

$$V_{\text{emf}} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

Motional EMF *loop moves, B static*

$$V_{\text{emf}} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$$

General (flux form) *always works*

$$V_{\text{emf}} = -\frac{d\Phi}{dt} \qquad \Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$$

N-turn coil *transformer, N turns*

$$V_{\text{emf}} = -N\frac{d\Phi}{dt} = -\frac{d\lambda}{dt} \qquad \lambda = N\Phi$$

Uniform B shortcut *B same everywhere on loop*

$$V_{\text{emf}} = -\frac{\partial B}{\partial t} \cdot A$$

Long wire + rect loop *wire near loop, r: $a \to b$, height c*

$$V_{\text{emf}} = -\int_0^c \int_a^b \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} = -\int_0^c \int_a^b \frac{\partial}{\partial t}\left(\frac{\mu_0 I}{2\pi r}\right) dr\, dz$$

MAXWELL'S EQUATIONS

Name	Differential form
Faraday	$\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$
Ampère	$\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D}/\partial t$
Gauss (E)	$\nabla \cdot \mathbf{D} = \rho_v$
Gauss (B)	$\nabla \cdot \mathbf{B} = 0$

$\mathbf{D} = \varepsilon\mathbf{E}$; $\mathbf{B} = \mu\mathbf{H}$; *source-free:* $\rho_v = 0$, $\mathbf{J} = \mathbf{0}$

SURFACE ELEMENT $d\mathbf{s}$

$d\mathbf{s} = \hat{n}\, da$ *(RHR gives $\hat{n}$: curl fingers in loop direction, thumb = $\hat{n}$)*

Loop plane	$d\mathbf{s}$
x-y	$\hat{z}\, dx\, dy$ or $\hat{z}\, r\, dr\, d\phi$
y-z	$\hat{x}\, dy\, dz$
x-z	$\hat{y}\, dx\, dz$
coplanar / toroid	$\hat{\phi}\, dr\, dz$

Uniform B: $\iint da = A$ (just the loop area)

DOT PRODUCTS $\hat{a} \cdot \hat{b}$

Same direction = 1; perpendicular = 0

	$\hat{x}$	$\hat{y}$	$\hat{z}$	$\hat{r}$	$\hat{\phi}$
$\hat{x}$	1	0	0	0	0
$\hat{y}$	0	1	0	0	0
$\hat{z}$	0	0	1	0	0
$\hat{\phi}$	0	0	0	0	1

CROSS PRODUCT $\mathbf{A} \times \mathbf{B}$ (determinant)

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

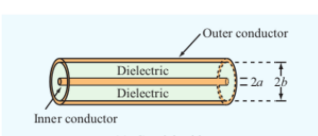
$= (A_yB_z - A_zB_y)\hat{x} - (A_xB_z - A_zB_x)\hat{y} + (A_xB_y - A_yB_x)\hat{z}$

Also used for curl:

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{vmatrix}$$

Plane wave ($\partial_x = \partial_y = 0$): only ∂_z row survives.

COAXIAL CABLE — AC RESISTANCE



(a) Coaxial cable

$$R' = \frac{R_s}{2\pi}\left(\frac{1}{a} + \frac{1}{b}\right) \quad (\Omega/\text{m})$$

a: inner radius; b: outer radius; $R_s = \sqrt{\pi f \mu/\sigma}$

B FIELDS

Infinite wire *single wire carrying I, $N = 1$*

$$\mathbf{B} = \hat{\phi}\,\frac{\mu_0 I}{2\pi r}$$

Inside toroid *coil of N turns carrying I*

$$\mathbf{B} = \hat{\phi}\,\frac{\mu N I}{2\pi r} \qquad \mu = \mu_r \mu_0$$

*N in $\mathbf{B}$ only if the **source** (current-carrying thing) is a multi-turn coil. Single wire $\Rightarrow$ $N = 1$, drop it. N in $V_{emf} = -N\,d\Phi/dt$ is the **receiver** coil turns.*

$\partial B/\partial t$ TABLE

$B(t)$	$\partial B/\partial t$
$B_0 \sin(\omega t)$	$B_0\omega\cos(\omega t)$
$B_0 \cos(\omega t)$	$-B_0\omega\sin(\omega t)$
$B_0 e^{-\alpha t}$	$-\alpha B_0 e^{-\alpha t}$
B_0 (const)	0

SIGN → CURRENT DIRECTION

V_{emf}	Inside source	I dir.
< 0	$- \to +$ terminal	$-\hat{\phi}$ (CW)
> 0	$+ \to -$ terminal	$+\hat{\phi}$ (CCW)
$= 0$	—	none

Lenz: induced I always opposes the change in Φ.

ln INTEGRAL

$B \propto 1/r$, loop spans $r = a$ to $r = b$

$$\int_a^b \frac{dr}{r} = \ln\frac{b}{a} = \ln b - \ln a$$

$\ln\frac{b}{a} = \ln b - \ln a$

$\ln 2 = 0.693 \quad \ln 3 = 1.099 \quad \ln 1.2 = 0.182$

ROTATING LOOP — GENERATOR

$$\Phi = AB_0\cos(\omega t + \phi_0)$$
$$V_{\text{emf}} = AB_0\omega\sin(\omega t + \phi_0)$$
$$I_{\text{amp}} = \frac{AB_0\omega}{R} \qquad \omega = 2\pi f = 2\pi\frac{\text{rpm}}{60}$$

Common loop areas

Shape	Area A
Square (side a)	a^2
Rectangle ($l \times w$)	lw
Circle (radius r)	πr^2
Equil. triangle (side a)	$\frac{\sqrt{3}}{4}a^2$
Any triangle	$\frac{1}{2}bh$

SI PREFIXES

Prefix	Sym	Value	Prefix	Sym	Value
pico	p	10^{-12}	kilo	k	10^3
nano	n	10^{-9}	mega	M	10^6
micro	μ	10^{-6}	giga	G	10^9
milli	m	10^{-3}	tera	T	10^{12}

j IDENTITIES

Expression	Result	Expression	Result
j^2	-1	$j \times (-j)$	1
$1/j$	$-j$	$-1/j$	j
j^3	$-j$	j^4	1
$e^{-j\pi/2}$	$-j$	$e^{j\pi/2}$	j
$e^{j\pi}$	-1	$e^{j2\pi}$	1

DISPLACEMENT & CONDUCTION CURRENT

Conduction *resistive, $\sigma \neq 0$*

$$E = \frac{V(t)}{d} \qquad J_c = \sigma E \qquad I_c = J_c \cdot A = \frac{\sigma A}{d}V(t)$$

$$I_c = \underbrace{\frac{\sigma A}{d}}_{1/R} V(t) \;\longleftrightarrow\; I = \frac{V}{R} \;\Rightarrow\; R = \frac{d}{\sigma A}$$

Current proportional to V directly $\to$ behaves like a resistor.

Displacement *capacitive, changing E*

$$J_d = \varepsilon\frac{\partial E}{\partial t} \qquad I_d = J_d \cdot A = \frac{\varepsilon_r\varepsilon_0 A}{d}\frac{dV}{dt}$$

$$I_d = \underbrace{\frac{\varepsilon_r\varepsilon_0 A}{d}}_{C}\frac{dV}{dt} \;\longleftrightarrow\; I = C\frac{dV}{dt} \;\Rightarrow\; C = \frac{\varepsilon_r\varepsilon_0 A}{d}$$

Current proportional to dV/dt $\to$ behaves like a capacitor.

Lossy cap circuit *both σ and ε_r present*

$R \parallel C$ across $V(t)$: $\quad I = I_c + I_d = \dfrac{V}{R} + C\dfrac{dV}{dt}$

CHARGE-CURRENT CONTINUITY

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

Conductor (steady): $\nabla \cdot \mathbf{J} = 0 \;\Rightarrow\; \oint_S \mathbf{J} \cdot d\mathbf{s} = 0$ *(KCL)*

CHARGE RELAXATION

Decay *charge injected at $t = 0$*

$$\rho_v(t) = \rho_{v0}\, e^{-t/\tau_r} \qquad \tau_r = \frac{\varepsilon_r\varepsilon_0}{\sigma}$$

Find σ from decay data

$$\sigma = \frac{-\ln(\rho_v/\rho_{v0}) \cdot \varepsilon_r\varepsilon_0}{t_1}$$

Larger τ_r = slower dissipation = better insulator.

PHASOR METHOD ($\mathbf{E} \to \mathbf{H}$)

1. Key relation $\quad \mathbf{E}(z,t) = \text{Re}\{\tilde{\mathbf{E}}(z)\, e^{j\omega t}\}$

2. Euler split $\quad e^{j(\omega t - kz)} = e^{j\omega t} \cdot e^{-jkz}$ $\quad$ strip $e^{j\omega t}$, keep e^{-jkz}

3. Time → phasor conversions

$\cos(\omega t - kz) = \text{Re}\{e^{j(\omega t - kz)}\} = \text{Re}\{e^{j\omega t} \cdot e^{-jkz}\} \to e^{-jkz}$

$\sin(\omega t - kz) = \cos(\omega t - kz - \frac{\pi}{2}) = \text{Re}\{e^{j(\omega t - kz - \pi/2)}\} = \text{Re}\{e^{-j\pi/2} \cdot e^{j\omega t} \cdot e^{-jkz}\} \to -j\, e^{-jkz}$

4. Faraday in phasor domain $\quad$ (plane wave: only $\partial/\partial z$ survives)

$$\tilde{\mathbf{H}} = -\frac{1}{j\omega\mu}\nabla \times \tilde{\mathbf{E}}$$

5. Phasor → time domain $\quad \mathbf{H}(z,t) = \text{Re}\{\tilde{\mathbf{H}}\, e^{j\omega t}\}$

$\text{Re}\{e^{j(\omega t - kz)}\} = \cos(\omega t - kz)$

$\text{Re}\{-j\, e^{j(\omega t - kz)}\} = \sin(\omega t - kz)$

UNIT CIRCLE

Angle	Deg	cos	sin	Q
0	0	1	0	—
$\pi/6$	30	$\sqrt{3}/2$	$1/2$	1
$\pi/4$	45	$\sqrt{2}/2$	$\sqrt{2}/2$	1
$\pi/3$	60	$1/2$	$\sqrt{3}/2$	1
$\pi/2$	90	0	1	—
$2\pi/3$	120	$-1/2$	$\sqrt{3}/2$	2
$3\pi/4$	135	$-\sqrt{2}/2$	$\sqrt{2}/2$	2
$5\pi/6$	150	$-\sqrt{3}/2$	$1/2$	2
π	180	-1	0	—
$7\pi/6$	210	$-\sqrt{3}/2$	$-1/2$	3
$5\pi/4$	225	$-\sqrt{2}/2$	$-\sqrt{2}/2$	3
$4\pi/3$	240	$-1/2$	$-\sqrt{3}/2$	3
$3\pi/2$	270	0	-1	—
$5\pi/3$	300	$1/2$	$-\sqrt{3}/2$	4
$7\pi/4$	315	$\sqrt{2}/2$	$-\sqrt{2}/2$	4
$11\pi/6$	330	$\sqrt{3}/2$	$-1/2$	4
2π	360	1	0	—

FIELDS

$\mathbf{E}$ — electric field (V/m)
$\mathbf{D} = \varepsilon\mathbf{E}$ — elec. flux density (C/m^2)
$\mathbf{B}$ — mag. flux density (T)
$\mathbf{H}$ — mag. field intensity (A/m)
ρ_v — charge density (C/m^3)
ρ_{v0} — initial charge density
$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s}$ — flux (Wb)
$\lambda = N\Phi$ — flux linkage (Wb)

CURRENTS

$J_c = \sigma E$ — cond. density (A/m^2)
$J_d = \varepsilon\, \partial E/\partial t$ — disp. density
$I_c = J_c \cdot A$ — cond. current (A)
$I_d = J_d \cdot A$ — disp. current (A)
$R = d/(\sigma A)$ — resistance (Ω)
$C = \varepsilon_r\varepsilon_0 A/d$ — capacitance (F)
$\tau_r = \varepsilon_r\varepsilon_0/\sigma$ — relax. time (s)

MATERIALS

$\mu_0 = 4\pi \times 10^{-7}$ H/m
μ_r — relative permeability
$\mu = \mu_r\mu_0$ — permeability
$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m
ε_r — relative permittivity
$\varepsilon = \varepsilon_r\varepsilon_0$ — permittivity
σ — conductivity (S/m)

CIRCUIT & WAVE

V — voltage (V)
R — resistance (Ω)
C — capacitance (F)
$\omega = 2\pi f$ — angular freq (rad/s)
$k = \omega\sqrt{\mu\varepsilon}$ — wave number (m^{-1})
$\lambda = 2\pi/k$ — wavelength (m)
$c = 3 \times 10^8$ m/s — speed of light

PHASOR & GEOMETRY

$\eta = \sqrt{\mu/\varepsilon}$ — intrinsic imp. (Ω)
$\eta_0 = 120\pi \approx 377\,\Omega$ (free space)
$k/(\omega\mu) = 1/\eta$
j — imaginary unit ($j^2 = -1$)
$\tilde{\mathbf{E}}$ — phasor (drop $e^{j\omega t}$)
A — area (m^2); $\quad d$ — sep. (m)
N — turns; $\quad r$ — radial dist. (m)

PLANE WAVE PARAMETERS (LOSSLESS)

Angular frequency *ω: rad/s; f: Hz*

$$\omega = 2\pi f \qquad f = \frac{\omega}{2\pi}$$

Phase velocity *m/s; nonmagnetic: $\mu_r = 1$*

$$u_p = \frac{c}{\sqrt{\varepsilon_r}} \qquad c = 3\times 10^8 \text{ m/s}$$

Wavenumber *k rad/m*

$$k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\varepsilon_r}}{c}$$

Wavelength & impedance *m & Ω*

$$\lambda = \frac{u_p}{f} = \frac{2\pi}{k}\ \text{(m)} \qquad \eta = \frac{\eta_0}{\sqrt{\varepsilon_r}}\ (\Omega),\ \eta_0 = 377\,\Omega$$

Order: $\omega \to u_p \to k \to \lambda \to \eta$. Base: $\mathbf{E} = E_0\cos(\omega t \pm kx + \phi_0)\hat{e}$.

LOSSLESS PLANE WAVE — E AND H

General E

$$\mathbf{E} = E_0\cos(\omega t - k\,\hat{k}\cdot\mathbf{r} + \phi_0)\,\hat{e}$$

$\hat{k}$ = direction of travel (unit vector), e.g. $+y \Rightarrow \hat{k} = \hat{y}$.
k = wavenumber (rad/m): how fast phase changes in space; $k = \omega\sqrt{\varepsilon_r}/c = 2\pi/\lambda$.
ϕ_0 = initial phase constant (rad): shift set by given condition at known t, position.
$\hat{e}$ = polarization direction of $\mathbf{E}$ (unit vector), e.g. E along $x \Rightarrow \hat{e} = \hat{x}$.
$\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, so $\hat{k}\cdot\mathbf{r}$ picks the axis, e.g. $\hat{k} = \hat{y} \Rightarrow \hat{k}\cdot\mathbf{r} = y$.

H from E *$\hat{k}$, $\hat{E}$, $\hat{H}$ form right-hand set*

$$\mathbf{H} = \frac{1}{\eta}(\hat{k}\times\mathbf{E})$$

Find ϕ_0 *given $E = E_1$ at $t = 0$, position r_0*

$$E_0\cos(-kr_0 + \phi_0) = E_1 \ \Rightarrow\ \phi_0 = \cos^{-1}\!\left(\frac{E_1}{E_0}\right) + kr_0$$

$\cos^{-1}$ gives $[0, \pi]$; take ϕ_0 in that range.

LOSSLESS vs. LOSSY — E & H

	Lossless ($\sigma = 0$)	Lossy ($\sigma \neq 0$)
Impedance	$\eta = \sqrt{\mu/\varepsilon}$	$\eta_c = \lvert\eta_c\rvert e^{j\theta_\eta}$
Prop.	e^{-jkz}	$e^{-\alpha z}e^{-j\beta z}$
$\mathbf{E}(t)$	$\mathrm{Re}\{\tilde{\mathbf{E}}e^{j\omega t}\}$ *(same)*	
$\mathbf{H}(t)$	$\mathrm{Re}\{\tilde{\mathbf{H}}\,e^{j\omega t}\}$ *(same)*	
$\tilde{\mathbf{H}}$ from $\tilde{\mathbf{E}}$	$\frac{1}{\eta}(\hat{k}\times\tilde{\mathbf{E}})$	$\frac{1}{\eta_c}(\hat{k}\times\tilde{\mathbf{E}})$
$\tilde{\mathbf{E}}$ from $\tilde{\mathbf{H}}$	$-\eta(\hat{k}\times\tilde{\mathbf{H}})$	$-\eta_c(\hat{k}\times\tilde{\mathbf{H}})$
Phase	in phase	H lags E by θ_η

Lossless shortcuts (wave in $+z$, $\mathbf{E}$ along $\hat{e}$):

$\mathbf{E}(z,t)$	$E_0\cos(\omega t - kz + \phi)\,\hat{e}$
$\mathbf{H}(z,t)$	$\frac{E_0}{\eta}\cos(\omega t - kz + \phi)\,(\hat{k}\times\hat{e})$
$\mathbf{H}(t)$ direct	$\frac{1}{\eta}(\hat{k}\times\mathbf{E}(t))$
$\mathbf{E}(t)$ direct	$-\eta(\hat{k}\times\mathbf{H}(t))$

DIRECTION OF PROPAGATION

Arg. contains	$\hat{k}$	Arg. contains	$\hat{k}$
$\omega t - kz$	$+\hat{z}$	$\omega t + kz$	$-\hat{z}$
$\omega t - ky$	$+\hat{y}$	$\omega t + ky$	$-\hat{y}$
$\omega t - kx$	$+\hat{x}$	$\omega t + kx$	$-\hat{x}$

Minus before k-axis $\Rightarrow$ + direction. Plus $\Rightarrow$ − direction.

$\hat{k}\times\hat{E}$ — FIND $\hat{H}$ DIRECTION

$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$-\hat{z}$	$\hat{x}$	$-\hat{y}$	$-\hat{z}$	$\hat{y}$	$+\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
$-\hat{y}$	$\hat{x}$	$+\hat{z}$	$-\hat{y}$	$\hat{z}$	$-\hat{x}$
$+\hat{x}$	$\hat{y}$	$+\hat{z}$	$+\hat{x}$	$\hat{z}$	$-\hat{y}$
$-\hat{x}$	$\hat{y}$	$-\hat{z}$	$-\hat{x}$	$\hat{z}$	$+\hat{y}$

Neg. $\hat{E}$: flip $\hat{H}$ sign. $\hat{k}\times\hat{k} = \mathbf{0}$; any $\hat{a}\times\hat{a} = \mathbf{0}$.
Cyclic: $\hat{x}\times\hat{y} = \hat{z}$, $\hat{y}\times\hat{z} = \hat{x}$, $\hat{z}\times\hat{x} = \hat{y}$. Reverse $\Rightarrow$ flip sign.

TRIG PHASE SHIFTS *($\theta \equiv \omega t - kz$)*

Input	Result	Input	Result
$\sin(\theta+\pi/2)$	$\cos\theta$	$\cos(\theta+\pi/2)$	$-\sin\theta$
$\sin(\theta-\pi/2)$	$-\cos\theta$	$\cos(\theta-\pi/2)$	$\sin\theta$
$\sin(\theta+\pi)$	$-\sin\theta$	$\cos(\theta+\pi)$	$-\cos\theta$
$\sin(\theta-\pi)$	$-\sin\theta$	$\cos(\theta-\pi)$	$-\cos\theta$

POLARIZATION — GENERAL WAVE

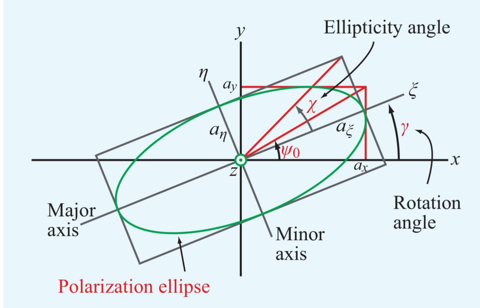


Figure 7-11 Polarization ellipse in the x–y plane with the wave traveling in the z direction (out of the page).

$$\mathbf{E} = a_x\cos(\omega t - kz)\hat{x} + a_y\cos(\omega t - kz + \delta)\hat{y}$$

a_x, a_y: bounding box half-widths (V/m); δ: phase diff (rad)
$\delta = 0$: $\mathbf{E} = (a_x\hat{x} + a_y\hat{y})\cos(\omega t - kz)$ (linear)
$\delta = \pi$: $\mathbf{E} = (a_x\hat{x} - a_y\hat{y})\cos(\omega t - kz)$ (linear)

Auxiliary angle *$\psi_0 = \arctan(a_y/a_x) \in (0, \pi/2)$*

Rotation angle γ (major axis from $\hat{x}$) *$\gamma \in [0, \pi)$*

$$\tan(2\gamma) = \tan(2\psi_0)\cos\delta$$

If sign(γ) $\neq$ sign(δ): $\gamma \leftarrow \gamma \pm \frac{\pi}{2}$.

Ellipticity angle χ *$\chi \in [-\pi/4, \pi/4]$*

$$\sin(2\chi) = \sin(2\psi_0)\sin\delta$$

Axial ratio $R = 1/\tan|\chi|$ *major axis = $R\times$ minor; $R \geq 1$*

χ SHAPE & HANDEDNESS

χ describes the shape of the ellipse and rotation direction.

χ (rad)	Polarization Type	Handedness
0	Linear	None
$(0,\ \pi/4)$	RHEP	RH (CCW)
$\pi/4$	RHC	RH (CCW)
$-\pi/4$	LHC	LH (CW)
$(-\pi/4,\ 0)$	LHEP	LH (CW)

QUADRANT RULE FOR $\tan(2\gamma)$

$N = \sin(2\psi_0)\cos\delta$, $\quad D = \cos(2\psi_0)$:

N	D	2γ (deg)	2γ (rad)
+	+	$\arctan(N/D)$	$\arctan(N/D)$
+	−	$180^\circ - \arctan\lvert N/D\rvert$	$\pi - \arctan\lvert N/D\rvert$
−	−	$180^\circ + \arctan\lvert N/D\rvert$	$\pi + \arctan\lvert N/D\rvert$
−	+	$360^\circ - \arctan\lvert N/D\rvert$	$2\pi - \arctan\lvert N/D\rvert$

$\gamma = (2\gamma)/2$. $\quad D = 0 \Rightarrow 2\gamma = \frac{\pi}{2} \Rightarrow \gamma = \frac{\pi}{4}$.

Convert: $\theta_{rad} = \theta_\circ \times \frac{\pi}{180}$; $\theta_\circ = \theta_{rad}\times\frac{180}{\pi}$.

PHASOR ↔ TIME DOMAIN

Phasor coeff	$\mathrm{Re}\{(\cdot)\,e^{j\omega t}e^{-\alpha z}e^{-j\beta z}\}$
$A\ \ (= Ae^{j\cdot 0})$	$A\cos(\omega t - \beta z)$
$-jA\ \ (= Ae^{-j\pi/2})$	$A\sin(\omega t - \beta z)$
$+jA\ \ (= Ae^{+j\pi/2})$	$-A\sin(\omega t - \beta z)$
$-A\ \ (= Ae^{\pm j\pi})$	$-A\cos(\omega t - \beta z)$
$Ae^{j\phi}$ (general)	$A\cos(\omega t - \beta z + \phi)$
$A + jB\ \ (= \lvert Z\rvert e^{j\theta})$	$\lvert Z\rvert\cos(\omega t - \beta z + \theta)$

$|Z| = \sqrt{A^2 + B^2}$, $\theta = \arctan(B/A)$ (check quadrant)

CIRCULAR POLARIZATION — TIME DOMAIN

$a_x = a_y = a$, wave in $+\hat{z}$

RHC *$\delta = -\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} - j\hat{y})e^{-jkz}$*

$$\mathbf{E}(z,t) = a\big[\cos(\omega t - kz)\,\hat{x} + \sin(\omega t - kz)\,\hat{y}\big]$$

LHC *$\delta = +\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} + j\hat{y})e^{-jkz}$*

$$\mathbf{E}(z,t) = a\big[\cos(\omega t - kz)\,\hat{x} - \sin(\omega t - kz)\,\hat{y}\big]$$

LOSSY MEDIA — CLASSIFICATION

Loss tangent

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0}$$

$\varepsilon''/\varepsilon'$	Type	Use
> 100	Good conductor	GC approx
0.01 to 100	Quasi-conductor	Exact eqs
< 0.01	Low-loss dielectric	LD approx

$\varepsilon' = \varepsilon_r\varepsilon_0$; $\varepsilon'' = \sigma/\omega$; nonmag.: $\mu = \mu_0$; free space: $\varepsilon_r = \mu_r = 1$.

ANY MEDIUM — EXACT ($X = \sqrt{1 + (\varepsilon''/\varepsilon')^2}$)

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X-1}\ \text{Np/m},\quad \beta = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X+1}\ \text{rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{u_p}{f}\ \text{(m)},\quad u_p = \frac{\omega}{\beta}\ \text{(m/s)}$$

$$\eta_c = \frac{\eta}{X^{1/2}}e^{j\theta_\eta}\ (\Omega),\quad \theta_\eta = \tfrac{1}{2}\arctan\frac{\sigma}{\omega\varepsilon'}\ \text{(rad)}$$

Rect: $\eta_c = |\eta_c|\cos\theta_\eta + j|\eta_c|\sin\theta_\eta$, $|\eta_c| = \eta/X^{1/2}$; $\eta = \eta_0/\sqrt{\varepsilon_r}$; $\varepsilon' = \varepsilon_r\varepsilon_0$.
Given $\tilde{\mathbf{H}} \propto e^{-\alpha y}e^{-j\beta y}$: $\gamma = \alpha + j\beta$; $\eta_c = j\omega\mu_0/\gamma = \omega\mu_0(\beta + j\alpha)/(\alpha^2 + \beta^2)$

SKIN DEPTH δ_s (m) — ANY LOSSY MEDIUM

General: $\delta_s = 1/\alpha$ *$|E(z)| = E_0e^{-z/\delta_s}$; lossless: $\alpha = 0 \Rightarrow$ no decay*
Low-loss ($\varepsilon''/\varepsilon' \ll 1$) *$\delta_s \approx 2/(\sigma\eta)$*
Good conductor ($\varepsilon''/\varepsilon' \gg 1$) *$\delta_s \approx 1/\sqrt{\pi f\mu\sigma}$*

LOW-LOSS MEDIUM ($\varepsilon''/\varepsilon' \ll 1$)

$$\alpha \approx \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon'}} = \frac{\sigma\eta}{2}\ \text{(Np/m)}$$

$$\beta \approx \omega\sqrt{\mu\varepsilon'} = \frac{\omega\sqrt{\varepsilon_r}}{c}\ \text{(rad/m)},\quad u_p \approx \frac{c}{\sqrt{\varepsilon_r}}\ \text{(m/s)}$$

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon'}} = \eta\ (\Omega,\ \angle \approx 0)$$

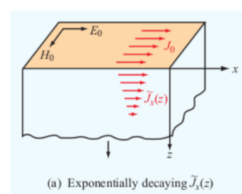
GOOD CONDUCTOR ($\varepsilon''/\varepsilon' \gg 1$)

$$\alpha \approx \beta \approx \sqrt{\pi f\mu\sigma}\quad \text{(Np/m, rad/m)}$$

$$u_p \approx \sqrt{4\pi f/\mu\sigma}\ \text{(m/s)},\quad \lambda \approx u_p/f\ \text{(m)}$$

$$\eta_c \approx (1+j)\sqrt{\frac{\pi f\mu}{\sigma}} \approx (1+j)\frac{\alpha}{\sigma}\quad (\Omega)$$

SURFACE RESISTANCE (GOOD CONDUCTOR)



$J_0 = \sigma E_0$ — current density at surface (z=0)
$\tilde{\mathbf{J}}(z) = J_0e^{-(1+j)z/\delta_s}\hat{x}$ — decays with depth
$R_s = \sqrt{\pi f\mu/\sigma} = 1/(\sigma\delta_s)$ (Ω)
$Z_s = (1+j)R_s$ — surface impedance
$R_{AC} = R_s\,\ell/w$ — AC resistance of slab

AVERAGE POWER DENSITY (POYNTING)

General *W/m²*

$$\mathbf{S}_{av} = \tfrac{1}{2}\mathrm{Re}\{\tilde{\mathbf{E}}\times\tilde{\mathbf{H}}^*\}$$

Lossless shortcut *$|E_0|^2 = a_x^2 + a_y^2 + a_z^2$*

$$\mathbf{S}_{av} = \frac{|E_0|^2}{2\eta}\hat{k}$$

Lossy medium *$\eta_c = |\eta_c|e^{j\theta_\eta}$; wave in $+\hat{z}$*

$$\mathbf{S}_{av} = \frac{|\tilde{\mathbf{E}}(0)|^2}{2|\eta_c|}e^{-2\alpha z}\cos\theta_\eta\ \hat{z}$$

Direction = $\hat{k}$. If wave goes in $-\hat{x}$, then $\mathbf{S}_{av}$ is a negative x-component.

DECIBELS & ATTENUATION

$G = 10\log(P_1/P_2) = 20\log(|E_1|/|E_2|)$ (dB)
$A = -8.69\,\alpha\,z$ (dB); α (dB/m) $\equiv 8.69\,\alpha$ (Np/m)

WAVE PARAMS

$\omega = 2\pi f$ — ang. freq (rad/s)
$k = \omega\sqrt{\varepsilon_r}/c$ — wavenumber (rad/m)
$\lambda = 2\pi/k$ — wavelength (m)
$u_p = c/\sqrt{\varepsilon_r}$ — phase vel. (m/s)
$\eta = \eta_0/\sqrt{\varepsilon_r}$ — impedance (Ω)
$c = 3\times 10^8$ m/s
$\eta_0 = 377\,\Omega$ (free space)

LOSSLESS MEDIA

$\sigma = 0$ (vacuum, air, ideal dielectric)
$\varepsilon'' = \sigma/\omega = 0 \ \Rightarrow\ \varepsilon = \varepsilon'$ (real)
$\alpha = 0$; no attenuation
$\eta = \eta_0/\sqrt{\varepsilon_r}$ real (Ω)
$\eta_0 = 377\,\Omega$ (free space)
E and **H** in phase ($\theta_\eta = 0$)

LOSSY MEDIA

α — atten. const (Np/m)
β — phase const (rad/m)
$\gamma = \alpha + j\beta$ — prop. const (m^{-1})
η_c — complex impedance (Ω)
$\varepsilon' = \varepsilon_r\varepsilon_0$; $\varepsilon'' = \sigma/\omega$ (F/m)
μ_r — rel. permeability ($\mu_r = 1$: nonmagnetic)
$\mu = \mu_r\mu_0$; $\mu_0 = 4\pi\times 10^{-7}$ H/m
$\delta_s = 1/\alpha$ — skin depth (m)

CIRCULAR WAVES

$\tilde{\mathbf{E}}_R = a_R(\hat{x} + e^{-j\pi/2}\hat{y})e^{-jkz} = a_R(\hat{x} - j\hat{y})e^{-jkz}$
$\tilde{\mathbf{E}}_L = a_L(\hat{x} + e^{+j\pi/2}\hat{y})e^{-jkz} = a_L(\hat{x} + j\hat{y})e^{-jkz}$
a_R, a_L, a_x in V/m; k in rad/m
Linear: $a_R = a_L = a_x/2$
$-j = e^{-j\pi/2}$: cos→sin
$+j = e^{+j\pi/2}$: cos→ −sin

USEFUL NUMBERS

$\sqrt{2.56} = 1.6$; $\sqrt{9} = 3$
$\sqrt{2.5} \approx 1.581$; $\sqrt{2} \approx 1.414$
$\arccos(2/3) \approx 48.2^\circ = 0.84$ rad
$\arctan(4/3) \approx 53.1^\circ = 0.927$ rad
$\arctan(3/4) \approx 36.9^\circ = 0.644$ rad
$\arctan(24/7) \approx 73.7^\circ = 1.287$ rad